Control Variates for a GARCH Model

Zero-Variance MCMC Estimation of the Posterior

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Outline

What is a Gaussian GARCH(1,1) Model?

- **Returns**: $\mathrm{r} = (r_1, \dots, r_T)$
- Conditional distribution:

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t)$$

- Variance recursion:

$$h_t = \omega_1 + \omega_2 r_{t-1}^2 + \omega_3 h_{t-1}$$

Definition: GARCH(1,1)

- **G**eneralized **A**uto**R**egressive
- **C**onditional **H**eteroskedasticity
- Models *time-varying volatility clusters*
- Variance depends on past residuals

Admissibility: $\omega_1 > 0,\ \omega_2 \geq 0,\ \omega_3 \geq 0,\ \omega_2 + \omega_3 < 1.$

Goal and Methodology

Estimate:

$$\mu_f = \mathbb{E}_\pi[f(X)]$$

where π is the target law (possibly unnormalised).

Core idea: Replace f by $\tilde{f}$ such that:

$$\mu_f = \mathbb{E}_\pi[\tilde{f}] \quad \text{and} \quad \mathbb{V}_\pi[\tilde{f}] < \mathbb{V}_\pi[f].$$

Variance Minimisation

A valid $\tilde{f}$ can be written as:

$$\tilde{f}(x) = f(x) + \frac{H\psi}{\sqrt{\pi(x)}}$$

where:

- H is a Hamiltonian operator with $H\sqrt{\pi} = 0$,
- ψ is smooth with compact support.

Zero-Variance Estimator and Approximation

Ideal case: If $\tilde{f} = \mu_f$, then $\sigma^2(\tilde{f}) = 0$.

This requires solving:

$$H\psi = -\sqrt{\pi(x)}\,[f(x) - \mu_f].$$

Approximation strategy:

1. Choose H such that $H\sqrt{\pi} = 0$;
2. Parametrize ψ;
3. Minimise $\sigma^2(\tilde{f})$ to derive optimal parameters;
4. Estimate parameters with a short MCMC;
5. Run a long MCMC using $\hat{\mu}_{\tilde{f}}$.

Choice of H and ψ

Choice of H

Schrödinger-type Hamiltonian:

$$Hf = -\frac{1}{2}\sum_{i=1}^{d}\frac{\partial^2}{\partial x_i^2}f + V(x)f.$$

Advantages:

- No closed-form conditional expectation needed;
- No reversibility assumption;
- Independent of $P(x, y)$ and MCMC algorithm.

Parametrize ψ to minimise $\sigma^2(\tilde{f})$.

The paper assumes:

$$\psi = P\sqrt{\pi},$$

where P is a polynomial.

Control Variate z

Combining H and ψ gives $\tilde{f}(x) = f(x) - \frac{1}{2}\Delta P(x) + \nabla P(x) \cdot z$.

The score-like control variate is:

$$z = -\frac{1}{2}\nabla \ln \pi(x), \quad \Delta = \sum_{i=1}^{d} \frac{\partial^2}{\partial x_i^2}.$$

The variance-reduction problem reduces to constructing z and choosing a polynomial basis P.

Why Metropolis–Hastings?

- Independent of the transition kernel $P(x, y)$;
- No reversibility required;
- ZV correction is a **post-processing** step.

Q1 — Random Walk Metropolis for GARCH

Goal: Implement a random walk Metropolis sampler for the GARCH posterior.

Data:

- Start with simulated data;
- Then use real data (e.g., log-returns from exchange rates).

Ingredients

- Prior on $(\omega_1, \omega_2, \omega_3)$ in the admissible region;
- Log-likelihood from h_t recursion;
- Gaussian random-walk proposal;
- Accept/reject step;
- Burn-in removal.

Q1 — Prior, Likelihood, and MH Steps

Prior and Likelihood

Prior: Truncated Gaussian on $\mathbb{R}_+^3$

$$\omega_k \sim \mathcal{N}_{\mathbb{R}_+}(0, s_k^2), \quad k = 1, 2, 3.$$

Log-likelihood:

$$\ln \mathcal{L}(r \mid \omega) = -\frac{1}{2} \sum_{t=1}^{T} \left[\ln h_t + \frac{r_t^2}{h_t} \right].$$

1. Initialise $\boldsymbol{\omega}^{(0)}$;
2. Propose $\boldsymbol{\omega}^\star = \boldsymbol{\omega}^{(i)} + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$;
3. Accept with probability:

$$\alpha = \min\left(1, \frac{\pi(\boldsymbol{\omega}^\star \mid r)}{\pi(\boldsymbol{\omega}^{(i)} \mid r)}\right);$$

4. Remove burn-in.

Q1 — Random walk Metropolis sampler using simulated data

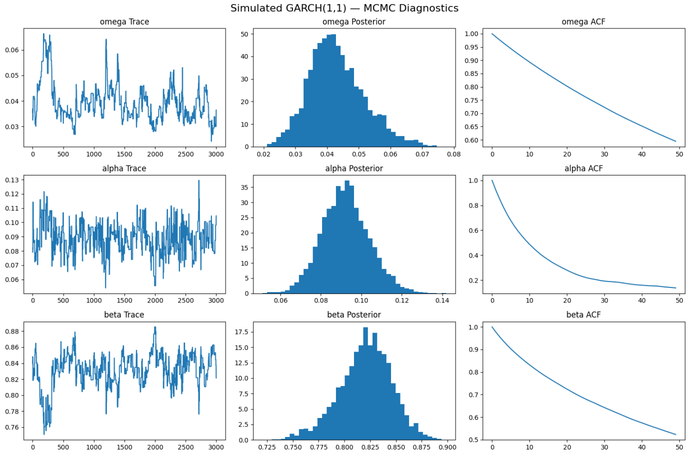

Figure: Using simulated data (true values: omega=0.05 alpha=0.1 beta=0.80)

Q1 — Random walk Metropolis sampler using real data

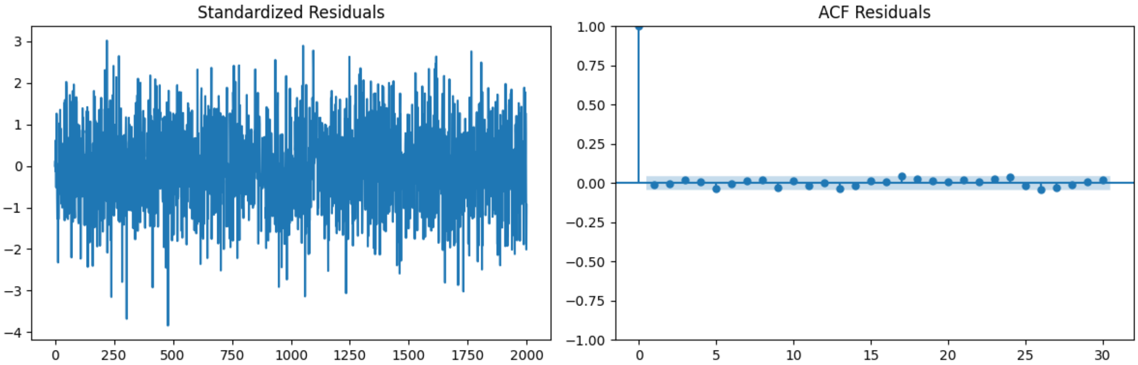

Figure: Residuals (Mean = -0.0043 Std = 0.9985)

Q1 — Random walk Metropolis sampler using real data

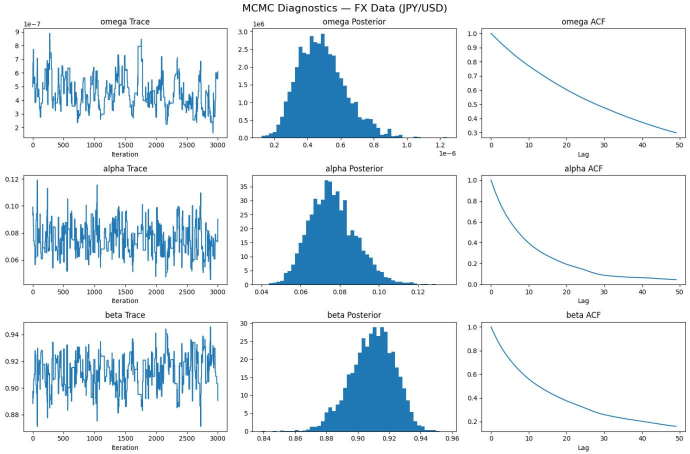

Figure: Using real data : exchange rates YEN/USD 2014-2025

Q1 — Random walk Metropolis sampler using real data

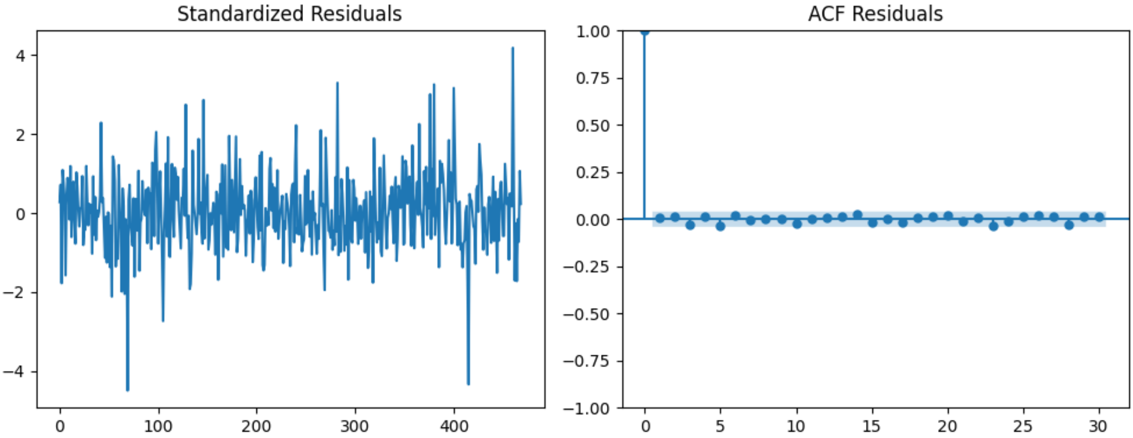

Figure: Residuals (Mean = 0.0153 Std = 0.9985)

Q1 — Random walk Metropolis sampler using real data

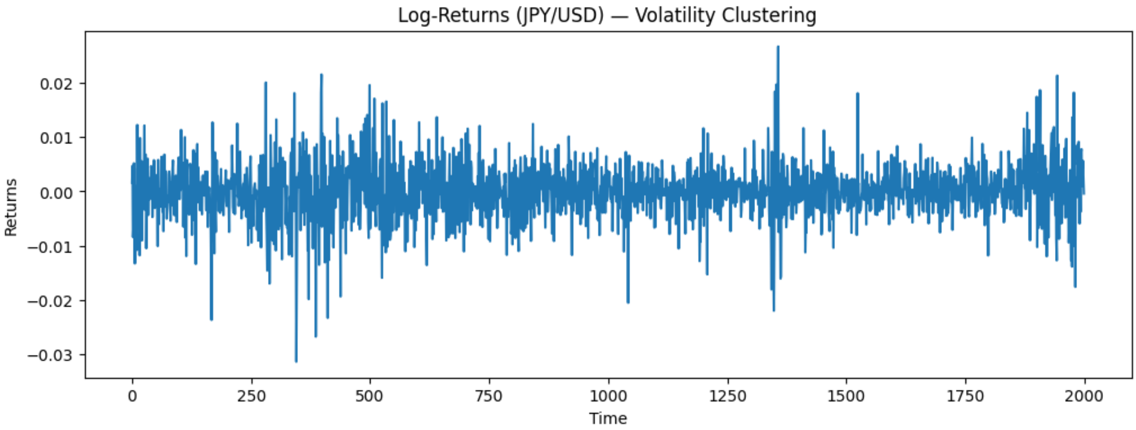

Figure: Log-returns JPY/USD

Q2 — Control Variates Setup

For $P(\omega) = a^\top \boldsymbol{\omega}$:

$$\tilde{f}(x) = f(x) + a^\top z(x),$$

where:

$$z = -\frac{1}{2}\nabla \log \pi(\theta).$$

Properties:

- $\mathbb{E}[f(x)] = \mathbb{E}[\tilde{f}(x)]$;
- $\mathbb{V}[\tilde{f}(x)] < \mathbb{V}[f(x)]$.

Variance Minimisation

Minimise:

$$\mathbb{V}[\tilde{f}(x)] = \mathbb{V}\big(f(x) - (-a^\top z(x))\big).$$

This reduces to:

$$\min_a \mathbb{E}[(f(x) - (-a^\top z(x)))^2].$$

This is a linear regression of $f(x)$ on $-z(x)$.

Q2 — OLS Implementation and z Properties

OLS Estimator: On a pilot sample $\{\omega^{(i)}\}_{i=1}^{N}$, estimate a by OLS and build:

$$\hat{\mu}_{\mathrm{ZV}} = \bar{f} - \hat{\mathrm{a}}^{\top} \bar{z}.$$

Why $\mathbb{E}_\pi[z] = 0$?

Definition

$$z(\omega) = -\frac{1}{2} \nabla \ln \pi(\omega).$$

Using $\nabla \ln \pi = \frac{\nabla \pi}{\pi}$:

$$z(\omega)\pi(\omega) = -\frac{1}{2} \nabla \pi(\omega).$$

$$\mathbb{E}_\pi[z] = \int z(\omega)\pi(\omega)\, d\omega = 0$$

(by integration by parts under boundary conditions).

Q2 — Calculation of z for GARCH

Posterior Density:

$$\pi(\omega \mid r) \propto \mathcal{L}(r \mid \omega)\pi_0(\omega)$$

$$\ln \pi(\omega \mid r) = -\frac{1}{2}\sum_{t=1}^{T}\left[\ln h_t + \frac{r_t^2}{h_t}\right] - \frac{1}{2}\sum_{k=1}^{3}\frac{\omega_k^2}{s_k^2} + c.$$

Expression for $z_j(\omega)$:

$$z_j(\omega) = -\frac{1}{2}\left[\sum_{t=1}^{T}\frac{1}{2h_t}\left(\frac{r_t^2}{h_t} - 1\right)\frac{\partial h_t}{\partial \omega_j} - \frac{\omega_j}{s_j^2}\right].$$

The only missing piece: recursive derivatives $\frac{\partial h_t}{\partial \omega_j}$.

Q2 — Recursive Derivatives of h_t

The derivatives satisfy:

$$\frac{\partial h_t}{\partial \omega_1} = 1 + \omega_3 \frac{\partial h_{t-1}}{\partial \omega_1}, \quad \frac{\partial h_t}{\partial \omega_2} = r_{t-1}^2 + \omega_3 \frac{\partial h_{t-1}}{\partial \omega_2}, \quad \frac{\partial h_t}{\partial \omega_3} = h_{t-1} + \omega_3 \frac{\partial h_{t-1}}{\partial \omega_3}.$$

Implications

- Derivatives can be computed **analytically** alongside h_t;
- No numerical differentiation needed;
- Control variates are fully analytical.

Q2 – Results (ZV-MCMC)

Parameter	Mean (Original)	Mean (ZV)	Var (Original)	Var (ZV)	Reduction
ω_1	0.1242	0.1235	3.35×10^{-4}	1.93×10^{-5}	17.35×
ω_2	0.1044	0.1036	8.10×10^{-5}	2.23×10^{-6}	36.30×
ω_3	0.7739	0.7752	5.75×10^{-4}	2.78×10^{-5}	20.70×

Control Variate Coefficients (β)

Parameter	β_1	β_2	β_3
ω_1	0.000484	0.000139	-0.000608
ω_2	0.000139	0.000164	-0.000279
ω_3	-0.000608	-0.000278	0.000870

Q2 – ZV-MCMC on Real Data (JPY/USD)

Parameter	Original Mean	ZV Mean	Original Var	ZV Var	Variance Reduction
ω_1	2.35×10^{-6}	3.12×10^{-11}	2.04×10^{-13}	1.09×10^{-40}	1.88×10^{27}
ω_2	1.21×10^{-1}	1.35×10^{-6}	6.69×10^{-7}	6.21×10^{-32}	1.08×10^{25}
ω_3	7.87×10^{-1}	8.69×10^{-6}	1.07×10^{-4}	2.82×10^{-30}	3.80×10^{25}

Analysis

- The variance reduction is extremely large for all parameters, suggesting that the control variates are highly effective in capturing the variability of the MCMC chain.
- However, the corrected variances are close to machine precision, which is suspicious and may indicate numerical issues.
- In addition, the ZV-corrected means are unexpectedly close to zero for all parameters, which is unlikely in practice and calls for caution.

Q2 – Optimized ZV-MCMC (Corrected Scaled Data)

Methodological Improvements

- **Data Scaling ($\times 100$):** Log-returns were multiplied by 100 to work with percentages. This prevents ω_1 from collapsing to 10^{-6}, avoiding machine precision errors and stabilizing gradients.
- **Adaptive Proposal:** Reduced *proposal_std* to 0.005 to better explore the high-density regions, accounting for the high persistence ($\omega_3 \approx 0.9$) found in JPY/USD volatility.

Parameter	Original Mean	ZV Mean	Var. Reduction
ω_1	0.004884	0.004958	4.42$\times$
ω_2	0.076651	0.078013	8.56$\times$
ω_3	0.909726	0.908398	5.11$\times$

Table: Results on JPY/USD (2343 obs)

Q3 — Larger Control Variate Basis

let's recall what we found $\tilde{f}(x) = f(x) - \frac{1}{2}\Delta P(x) + \nabla P(x) \cdot z.$ (8)

where:

$$z = -\frac{1}{2}\nabla \log \pi(\theta).$$

Degree-2 Polynomial Basis: Let us recall that the whole method works by choosing a class of polynomials P, then estimating their coefficients so that the variance of $\tilde{f}$ is as small as possible. Increasing the degree of P means expanding the family of control variates. That is exactly where the computational issue appears :

- For $d = 3$, since it is a GARCH(1,1) we have:$x = \omega = (\omega_1, \omega_2, \omega_3)$,
- P of degree 2 : $P(x) = c + a^\top x + \frac{1}{2}x^\top Bx$

Knowing that $\nabla P(x) = a + Bx, \qquad \Delta P(x) = \mathrm{tr}(B)$

We have $\tilde{f}(x) = f(x) - \frac{1}{2}\,\mathrm{tr}(B) + (a + Bx)^\top z(x).$

Q3 — Larger Control Variate Basis

Expanding this expression, we have :

$$\tilde{f}(x) = f(x) + \sum_{j=1}^{d} a_j z_j(x) + \sum_{j=1}^{d} b_{jj}\left(x_j z_j(x) - \frac{1}{2}\right) + \sum_{1 \le j < k \le d} b_{jk}\left(x_j z_k(x) + x_k z_j(x)\right).(9)$$

In this case,

$$z(\omega) = -\frac{1}{2}\nabla_\omega \log \pi(\omega) = \left(z_1(\omega), z_2(\omega), z_3(\omega)\right).$$

If $P(\omega)$ is quadratic, the family of control variates generated by (8) is therefore

$$h(\omega) = \Big(z_1(\omega),\ z_2(\omega),\ z_3(\omega),\ \omega_1 z_1(\omega) - \frac{1}{2},\ \omega_2 z_2(\omega) - \frac{1}{2},\ \omega_3 z_3(\omega) - \frac{1}{2},$$

$$\omega_1 z_2(\omega) + \omega_2 z_1(\omega),\ \omega_1 z_3(\omega) + \omega_3 z_1(\omega),\ \omega_2 z_3(\omega) + \omega_3 z_2(\omega)\Big)^\top.$$

There are exactly 9 such functions, because

$$\frac{d(d+3)}{2} = \frac{3(3+3)}{2} = 9.$$

Q3 — Larger Control Variate Basis

As this control-variate framework can be viewed as a regression problem, when the control space becomes richer, the approximation may improve, but two practical problems arise:

- The regression design matrix may become highly collinear;
- The OLS computation may become expensive

Why can this become expensive? Because OLS on m control variates requires forming and inverting the empirical Gram matrix

$$\mathbb{E}_\pi\left[h(\omega)h(\omega)^\top\right]$$

which is of size 9×9.
The paper states that the computational cost of OLS is of order

$$O(n9^2 + 9^3),$$

In general by recalling that number of variable id $\frac{d(d+3)}{2}$, the OLS cost scales as (only for a 2 degree Polynomial !!)

$$O(nd^4 + d^6).$$

This is why we now will move from plain OLS to regularized methods such as Lasso.

Q3 — Lasso-Based Adaptation

Take the first Nn_0 draws of the pilot chain and use the same centered quantities, but compute on these N draws. Then define the Lasso estimator of the coefficients of P(x) as the minimizer of :

Lasso Estimator

Replace OLS with:

$$\frac{1}{2N}\sum_{i=1}^{N}\Bigg[\left(f(x^{(i)})-\bar{f}_N\right)+\sum_{j=1}^{d}a_j\left(z_j(x^{(i)})-\bar{z}_{j,N}\right)+\sum_{j=1}^{d}b_{jj}\left(x_j^{(i)}z_j(x^{(i)})-\frac{1}{2}-\overline{x_jz_j-\frac{1}{2}}_N\right)$$

$$+\sum_{1\le j<k\le d}b_{jk}\left(x_j^{(i)}z_k(x^{(i)})+x_k^{(i)}z_j(x^{(i)})-\overline{x_jz_k+x_kz_j}_N\right)\Bigg]^2+\lambda\left(\sum_{j=1}^{d}|a_j|+\sum_{j=1}^{d}|b_{jj}|+\sum_{1\le j<k\le d}|b_{jk}|\right).$$

. **Why Lasso ?**

- it penalizes the coefficients of the polynomial P(x);
- if one of these coefficients is set to zero, then the corresponding control variate is removed;
- selects a sparse quadratic polynomial P(x), and at the same time a sparse family of control variates.

Q3 — Post-Lasso

Post-Lasso step :
As in Leluc, Portier and Segers, we should not use the penalized coefficients as final coefficients. After the Lasso has selected the nonzero terms, we refits ordinary least squares on that selected set only. In words :

- the Lasso chooses which coefficients of P(x) are allowed to be nonzero;
- then OLS estimates those selected coefficients without penalty.

We get $\left(\hat{a}^{\mathrm{PL}}, \hat{B}^{\mathrm{PL}}\right)$ the minimizer of the same least-squares criterion as in (9), but under the constraint from the lasso

Q3 — Plug-in Estimator

Once $(\hat{a}^{\mathrm{PL}}, \hat{B}^{\mathrm{PL}})$ have been obtained from the pilot run, define the final controlled function by plugging them into equation (9):

$$\tilde{f}_{\mathrm{PL}}(x) = f(x) - \frac{1}{2}\operatorname{tr}\big(\hat{B}^{\mathrm{PL}}\big) + \big(\hat{a}^{\mathrm{PL}} + \hat{B}^{\mathrm{PL}}x\big)^{\top} z(x).$$

Then, on a second long MCMC run

$$x_*^{(1)}, \ldots, x_*^{(n_1)},$$

the final estimator is

$$\hat{\mu}_f^{\mathrm{PL}} = \frac{1}{n_1}\sum_{i=1}^{n_1} \tilde{f}_{\mathrm{PL}}\big(x_*^{(i)}\big).$$

Equivalently,

$$\hat{\mu}_f^{\mathrm{PL}} = \frac{1}{n_1}\sum_{i=1}^{n_1}\left[f\big(x_*^{(i)}\big) - \frac{1}{2}\operatorname{tr}\big(\hat{B}^{\mathrm{PL}}\big) + \big(\hat{a}^{\mathrm{PL}} + \hat{B}^{\mathrm{PL}}x_*^{(i)}\big)^{\top} z\big(x_*^{(i)}\big)\right].$$

So the final estimator still has exactly the ZV form of Mira et al.; the only change is how the coefficients of $P(x)$ are estimated.

Q3 — Comparison: Naive vs. Lasso

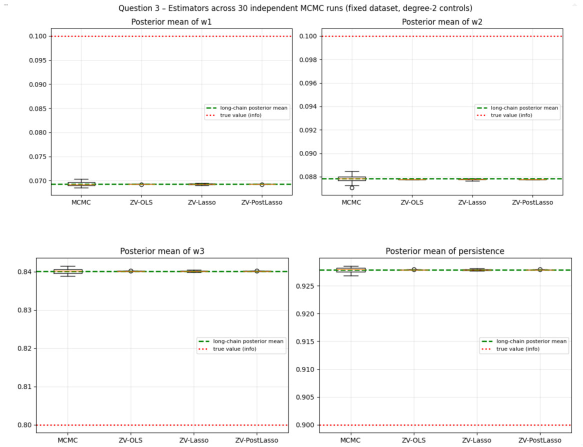

Figure: Estimators across 30 independent MCMC runs

Q3 — Comparison: Naive vs. Lasso

Variance reduction. Degree-2 ZV estimators are much more concentrated than raw MCMC: the boxplots are dramatically narrower, with variance-reduction factors of about $200\times$ to $1400\times$. This confirms that enlarging the control space from degree 1 to degree 2 yields a major gain in Monte Carlo precision.

Bias. Relative to the long-chain posterior mean (green line), all estimators remain well centered, so there is no visible additional bias. In this GARCH(1,1) example, the degree-2 basis contains only 9 controls, which explains why plain ZV-OLS already performs very well.

OLS vs Lasso. OLS, Lasso and Post-Lasso give very similar point estimates, but OLS usually has the smallest spread. This is expected: Lasso shrinks some coefficients toward zero, weakening part of the control-variate correction, while Post-Lasso recovers most of it through an OLS refit; in this small 9-control setting, regularization brings no clear gain and would be more useful for much larger control dictionaries.

In Conclusion : There is no real gain using a Lasso for a 2-degree Polynomial.

Q4 — Is OLS Valid on MCMC Samples?

Short answer: No, because MCMC samples are **dependent**, while OLS assumes independence.

Consequences:

- Consecutive draws are autocorrelated;
- Effective sample size $< N$;
- OLS point estimates may still work, but standard errors are unreliable.

What Remains Valid

Under ergodicity, empirical averages converge. The ZV estimator can still reduce variance.

Q4 — Remedies for Dependence

Sub-sampling

Keep one draw every k iterations.

Pros: Reduces autocorrelation.

Cons: Wastes data.

Block Averages

Average over non-overlapping blocks.

Pros: Reflects long-run variability.

Independent Chains

Run K short independent chains and average their ZV estimates.

Pros: Truly i.i.d. between chains; unbiased combination.

Cons: Requires K pilot runs; less efficient than one long chain.

Key idea: Always think in terms of **effective sample size**.

Q4 — Comparison of three strategies to restore (approximate) independence

Method	Est. $\mathbb{E}[\omega_1]$	N used	ESS
True value	0.10000	—	—
Raw MC mean	0.11821	10000	791
ZV-OLS (raw, autocorr.)	0.11899	10000	—
ZV-OLS thin = 5	0.11900	2000	744
ZV-OLS thin = 10	0.11898	1000	635
ZV block means ($m = 50$)	0.11899	200	146
ZV block means ($m = 100$)	0.11899	100	89
ZV independent chains ($K = 10$)	0.11895	10	i.i.d.

Table: Chain length $N = 10000$, ESS $= 791$, IAT ≈ 12.6

Q4 — Comparison of three strategies to restore (approximate) independence

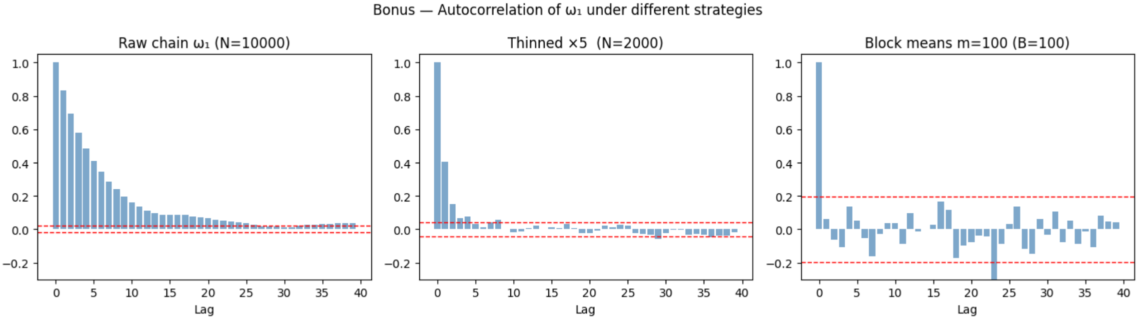

Figure: Autocorrelation under different strategies

Q4 — Conclusion

- The ZV point estimate is consistent regardless of autocorrelation (ergodic theorem).
- Thinning and block means restore approximate independence but discard data.
- Independent chains provide truly i.i.d. ZV estimates at the cost of K pilot runs.
- In practice, raw ZV-OLS on the full chain is already the best point estimator; the strategies above matter most when reporting confidence intervals for $\hat{\beta}$.

Q4 — Practical Takeaways

In practice:

- Control variates can still be estimated from MCMC samples;
- Variance reduction is possible;
- But inference must account for dependence;
- Use effective sample size, not raw N.

The zero-variance idea remains useful, but dependence must be addressed for uncertainty quantification.

Summary

1. GARCH(1,1) provides a natural setting with analytically tractable score derivatives.
2. Zero-variance control variates replace f by $\tilde{f}$ with same expectation and smaller variance.
3. With degree-1 polynomials, coefficient estimation reduces to linear regression of f on $-z$.
4. The control variate $z = -\frac{1}{2}\nabla \log \pi$ is computed analytically from the GARCH recursion.
5. For larger bases, Lasso avoids OLS instability.
6. MCMC dependence requires adjusted inference (thinning, blocking, independent chains).

Thank you for your attention

Questions?